
NYCU IMF - 114 Financial Econometrics

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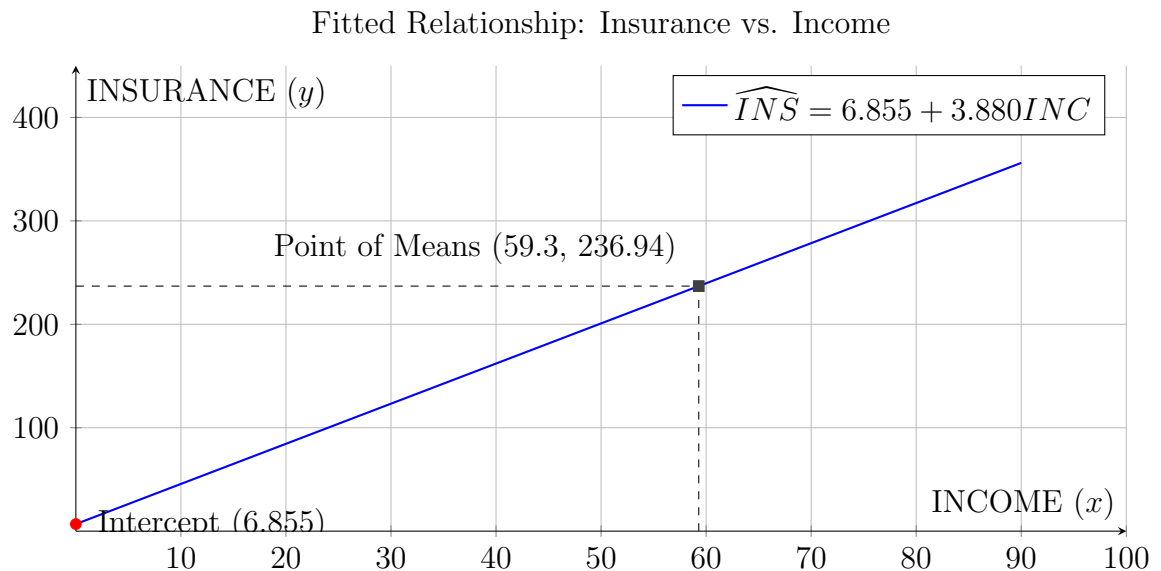
Due Date: March 24, 2026
Assignment: 2, Ch3 q18, 23, 31

Question 18

A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\begin{array}{rcc} \widehat{INSURANCE} & = & 6.855 + 3.880 INCOME \\ (se) & & (7.383) \quad (0.112) \end{array}$$

- (a) Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.



- (b) How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

According to the provided relationship between *INSURANCE* and *INCOME*, we estimate that the average amount of insurance held changes is positive 3.880 thousand dollars with each additional \$1000 of household income. The point estimate is 3.880.

And to find the 95% confidence interval for the slope coefficient (β_2), we use the formula:

$$b_2 \pm t_c \cdot se(b_2)$$

- Point Estimate (b_2): 3.880
- Standard Error ($se(b_2)$): 0.112 (given)
- Degrees of Freedom (df): $N - 2 = 20 - 2 = 18$
- Critical Value (t_c): For a 95% confidence level ($\alpha = 0.05$, two-tailed) with 18 degrees of freedom, the critical t -value is 2.101.

Plugging these values into the formula:

$$3.880 \pm 2.101(0.112)$$

$$3.880 \pm 0.2353$$

This gives us an interval estimate of:

$$[3.6447, 4.1153]$$

Explanation to Stockholders:

We are 95% confident that for every additional \$1,000 a household earns in income, the average amount of life insurance they purchase increases by somewhere between \$3,645 and \$4,115.

- (c) Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746.

Step 1: Calculate the Point Estimate ($\hat{y}_0$)

First, we calculate the expected amount of insurance for a household with a \$100,000 income. Since the *INCOME* variable (x) is in thousands of dollars, we plug in $X_0 = 100$.

$$\hat{y}_0 = b_1 + b_2 X_0$$

$$\hat{y}_0 = 6.855 + 3.880(100)$$

$$\hat{y}_0 = 394.855$$

So, the point estimate is \$394.855 (or \$394,855).

Step 2: Calculate the Variance of the Estimate

To construct the confidence interval, we need the standard error of this expected value, denoted as $se(\hat{y}_0)$. The formula for the estimated variance of the expected value is:

$$\widehat{var}(\hat{y}_0) = \widehat{var}(b_1) + X_0^2 \widehat{var}(b_2) + 2X_0 \widehat{cov}(b_1, b_2)$$

We can find these components using the information provided in the problem:

- $\widehat{var}(b_1)$ is the square of the standard error of the intercept: $7.383^2 = 54.5087$
- $\widehat{var}(b_2)$ is the square of the standard error of the slope: $0.112^2 = 0.012544$
- $X_0 = 100$
- $\widehat{cov}(b_1, b_2) = -0.746$ (given in part c)

Then, we plug them into the variance formula:

$$\widehat{var}(\hat{y}_0) = 54.5087 + (100^2 \cdot 0.012544) + 2(100)(-0.746)$$

$$\widehat{var}(\hat{y}_0) = 54.5087 + 10000(0.012544) - 149.2$$

$$\widehat{var}(\hat{y}_0) = 54.5087 + 125.44 - 149.2$$

$$\widehat{var}(\hat{y}_0) = 30.7487$$

The standard error is the square root of the variance:

$$se(\hat{y}_0) = \sqrt{30.7487} = 5.545$$

Step 3: Find the Critical t -value

We need a 99% interval estimate.

- Degrees of Freedom (df): $N - 2 = 20 - 2 = 18$
- Confidence Level: 99

Searching at a t -distribution table for $df = 18$ and a two-tailed area of 0.01, the critical t -value (t_c) is 2.878.

Step 4: Construct the 99% Interval Estimate

The confidence interval of the expected value is:

$$\begin{aligned} & \hat{y}_0 \pm t_c \cdot se(\hat{y}_0) \\ & 394.855 \pm 2.878(5.545) \\ & 394.855 \pm 15.958 \\ & [378.897, 410.813] \end{aligned}$$

The 99% interval estimate for the expected amount of insurance held by a household with a \$100,000 income is [378.897, 410.813] (in thousands of dollars), which translates to between \$378,897 and \$410,813

- (d) One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic

used and the rejection region.

From the board member's claim into our model's units, since both *INCOME* and *INSURANCE* are measured in thousands of dollars, a \$5,000 increase in insurance means the *INSURANCE* variable increases by 5. Therefore, the claim is stating that the true slope parameter (β_2) is exactly 5.

1. The Hypotheses

To test if the claim is true against the alternative that it is not, which makes this a two-tailed test.

- Null Hypothesis (H_0): $\beta_2 = 5$ (The claim is true)
- Alternative Hypothesis (H_1): $\beta_2 \neq 5$ (The claim is not true)

2. The test statistic

The test statistic for a single parameter follows a t -distribution. We use the point estimate (b_2), the hypothesized value (β_2), and the standard error of the slope ($se(b_2)$).

$$\begin{aligned}t &= \frac{b_2 - \beta_2}{se(b_2)} \\t &= \frac{3.880 - 5}{0.112} \\t &= \frac{-1.120}{0.112} \\t &= -10\end{aligned}$$

3. The Rejection Region

We need the critical t -values for a two-tailed test at a 5% level of significance ($\alpha = 0.05$).

- Degrees of Freedom (df): $N - 2 = 20 - 2 = 18$
- Critical t -value (t_c): For $\alpha/2 = 0.025$ and $df = 18$, the critical value is 2.100922.

The rejection region is defined as $t \leq -2.100922$ or $t \geq 2.100922$.

4. The conclusion

Our calculated test statistic is $t = -10$. Since $-10 < -2.100922$, it falls squarely within the rejection region. Therefore, we reject the null hypothesis at the 5% significance level. The data do not support the management board member's claim; the actual increase is significantly different from (and statistically lower than) \$5,000 per \$1,000 of income.

- (e) Test the hypothesis that as income increases, the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of

significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

The problem asks whether the amount of life insurance increases by the same amount as income, and both variables are measured in thousands of dollars. Therefore, we are testing if the slope parameter (β_2) is exactly equal to 1.

1. The Hypotheses

As the problem explicitly asks us to use an alternative hypothesis that the slope is larger than one, making this a one-tailed (right-sided) test.

- Null Hypothesis (H_0): $\beta_2 = 1$
- Alternative Hypothesis (H_1): $\beta_2 > 1$

2. Test Statistic

Use the same t -statistic formula as in part d, but substitute a new hypothesized value.

$$\begin{aligned}t &= \frac{b_2 - \beta_2}{se(b_2)} \\t &= \frac{3.880 - 1}{0.112} \\t &= \frac{2.880}{0.112} \\t &\approx 25.714286\end{aligned}$$

3. The Reject Region

We need the critical t -value for a one-tailed test at a 1% level of significance ($\alpha = 0.01$).

- Degrees of Freedom (df): $N - 2 = 20 - 2 = 18$
- Critical t -value (t_c): Looking up $\alpha = 0.01$ for a one-tailed test with 18 degrees of freedom, the critical value is 2.55238.

The rejection region is defined as $t \geq 2.55238$

4. Conclusion

Our calculated test statistic is $t = 25.714$. Because 25.714 is significantly greater than our critical value of 2.55238, it falls deep into the rejection region. Therefore, we reject the null hypothesis at the 1% level of significance.

We have strong statistical evidence to conclude that as household income increases, the amount of life insurance held increases by a significantly larger amount than the income increase itself (i.e., the true slope is statistically greater than one).

Question 23

The data file *college town* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

- (a) Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

The marginal effect of a 1-unit increase in *SQFT* (100 square feet) for a house with 2000 square feet ($SQFT = 20$) is given by the derivative evaluated at $SQFT = 20$:

$$\frac{dE(PRICE)}{dSQFT} = 2\alpha_2(20) = 40\alpha_2$$

We test the following hypotheses at the $\alpha = 0.05$ level:

$$H_0 : 40\alpha_2 \leq 13$$

$$H_1 : 40\alpha_2 > 13$$

The test statistic is calculated as:

$$t = \frac{40\hat{\alpha}_2 - 13}{40 \cdot se(\hat{\alpha}_2)} = \frac{40(0.184519) - 13}{40(0.005256)} = \frac{7.381 - 13}{0.210} = -26.728$$

For $df = 498$ and $\alpha = 0.05$ (one-tailed), the critical value is 1.647919. The rejection region is $t \geq 1.647919$.

The calculated p -value is roughly 1.000. Because the test statistic ($t = -26.728$) does not fall into the rejection region (and the p -value is > 0.05), we fail to reject the null hypothesis. We conclude that there is no statistical evidence that the marginal effect of adding 100 square feet to a 2000 square foot house is greater than \$13,000.

- (b) Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

For a house with 4000 square feet ($SQFT = 40$), the marginal effect of an additional 100 square feet is:

$$\frac{dE(PRICE)}{dSQFT} = 2\alpha_2(40) = 80\alpha_2$$

We test the hypotheses at the $\alpha = 0.05$ level:

$$H_0 : 80\alpha_2 \leq 13$$

$$H_1 : 80\alpha_2 > 13$$

Using the regression output, the test statistic is:

$$t = \frac{80\hat{\alpha}_2 - 13}{80 \cdot se(\hat{\alpha}_2)} = \frac{80(0.184519) - 13}{80(0.005256)} = \frac{14.762 - 13}{0.420} = 4.189$$

The critical value remains 1.647919 for $df = 498$ and $\alpha = 0.05$, and the rejection region is $t \geq 1.647919$.

Because our test statistic ($t = 4.189$) falls well within the rejection region, we reject the null hypothesis. We conclude that for a 4000 square foot house, the marginal effect of an additional 100 square feet on the expected price is statistically significantly greater than \$13,000.

- (c) Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

To estimate the expected price for a house of 2000 square feet, we set $SQFT = 20$. Using the estimated quadratic regression model, the point estimate is:

$$\begin{aligned} E(PRICE|\widehat{SQFT} = 20) &= \hat{\alpha}_1 + \hat{\alpha}_2(20^2) \\ &= 93.566 + 0.1845(400) \\ &= 167.3735 \end{aligned}$$

The expected selling price is approximately \$167,374.

The 95% interval estimate for the expected price is calculated as:

$$[158.0481, 176.6988]$$

General Audience Explanation:

We are 95% confident that the average selling price for all 2,000 square foot houses in this specific real estate market is between \$158,048 and \$176,699.

- (d) Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

First, we locate houses in the sample with exactly 2000 square feet of living area ($SQFT = 20$). There are exactly 3 houses in the dataset that meet this criterion.

The sample mean of their selling prices is:

$$\bar{PRICE}_{SQFT=20} = 163.3333$$

This translates to an average selling price of approximately \$163,333.

To determine if this is compatible with the result in part (c), we compare it to our model's point estimate of 167.374 and its corresponding 95% interval estimate. Because 163.3333 is very close to our predicted expected value, it easily falls within the 95% confidence interval. Therefore, we conclude that the sample average is highly compatible with the expected price estimated by the quadratic regression model.

Explanation: The regression model utilizes information from all 500 observations to estimate the expected value, smoothing out random variation to capture the broader market trend. The sample mean, however, relies solely on a very small subset of 3 observations. Given this tiny sample size, finding a sample mean (\$163,333) that is this close to the model's prediction (\$167,374) demonstrates strong compatibility between the raw data and our fitted quadratic model.

Question 31

Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable *SAL1* = unit sales of brand no. 1 canned tuna, and *APR1* = price per can of brand no. 1 tuna (in dollars).

- (a) Calculate the summary statistics for *SAL1* and *APR1*. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus *WEEK*. How much variation in sales and price is there from week to week?

The summary statistics for the weekly sales (*SAL1*) and price (*APR1*) of brand no. 1 canned tuna are as follows:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1762	2485	3136	6719	8612	32820

Table 1: SAL1 Summary statistics

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6100	0.6900	0.8100	0.7825	0.8625	0.9200

Table 2: APR1 Summary statistics

According to the summary statistic, for the *SAL1*, the mean is 6719, the minimum is 1762, the maximum is 32820, and have a standard deviations 6915.083. And for the *APR1*, the mean is 0.7825, the minimum is 0.6100, the maximum is 0.9200, and have a standard deviations 0.09803711.

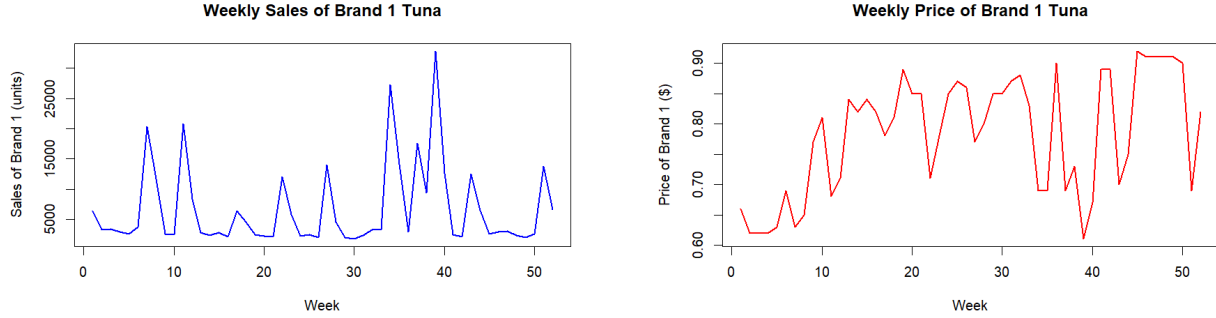


Figure 1: Weekly Variation in Sales and Price

Extent of Variation:

To quantify the week-to-week variation, we look at the standard deviation relative to the mean, known as the Coefficient of Variation ($CV = \sigma/\mu$).

- **Sales (*SAL1*):** The CV is 102.9228%, indicating highly volatile week-to-week sales.
- **Price (*APR1*):** The CV is 12.5287%, indicating moderate week-to-week price fluctuations.

Visually, the time series plots confirm this. The price plot shows a relatively stable baseline with occasional, sharp downward spikes (promotional discounts). Conversely, the sales plot exhibits massive upward spikes that perfectly mirror those price drops, demonstrating that a relatively small percentage variation in price triggers a massive percentage variation in sales volume.

- (b) Plot the variable *SAL1*(y-axis) against *APR1*(x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?

Based on the scatterplot in Fig 2, there is a clear inverse (negative) relationship between *SAL1* and *APR1*. As the price of the tuna increases, the unit sales decrease. This is exactly what we expect based on basic economic theory (the Law of Demand). Canned tuna is a normal good, and as it becomes more expensive, consumers will buy less of it, either choosing to substitute it with a cheaper competing brand or opting for a different type of food entirely.

- (c) Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in

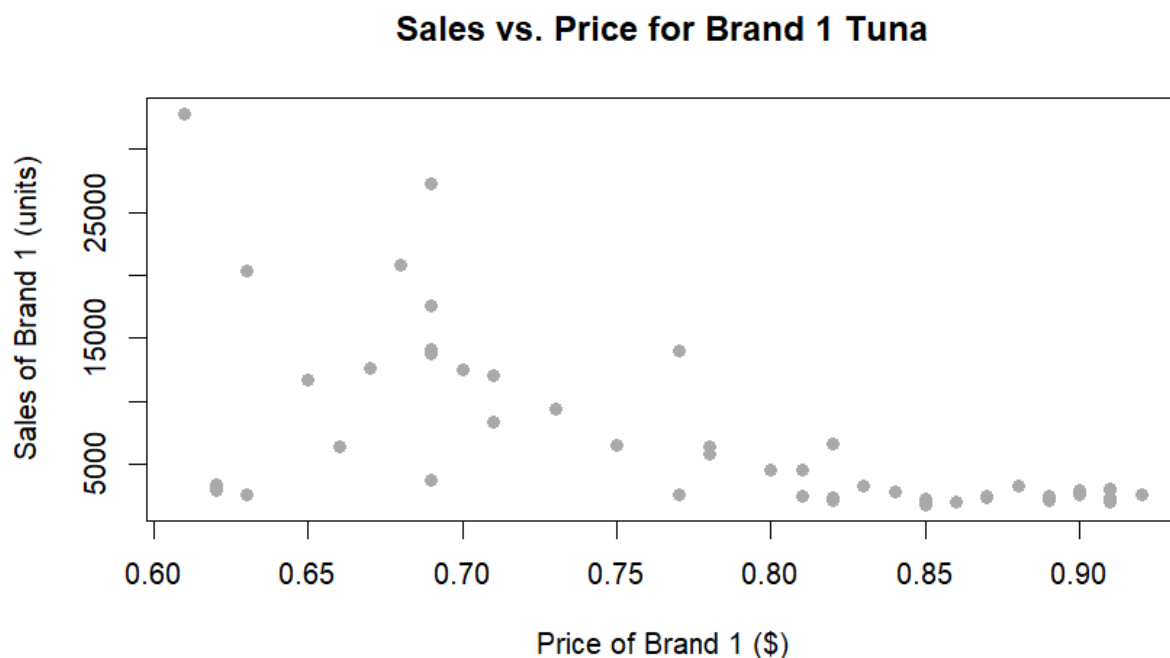


Figure 2: Scatterplot of Sales versus Price

the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?

First, we create the new variable $PRICE1 = 100 \times APR1$ to represent the price in cents. We then estimate the linear regression model:

$$\widehat{SAL1} = \beta_1 + \beta_2 PRICE1$$

From the regression output, the point estimate for the effect of a one-cent increase in the price of brand no. 1 tuna on its sales is the estimated slope coefficient:

$$b_2 = -434.45$$

This means that, on average, a one-cent increase in price is associated with a decrease of approximately 434.45 units sold per week.

Using the standard error of the estimate, the 95% interval estimate for this effect is:

$$[-592.2897, -276.6049]$$

We are 95% confident that a one-cent increase in the price of brand no. 1 tuna decreases the expected weekly sales by an amount between 276.6049 and 592.2897 units.

- (d) Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.

To estimate the expected number of cans sold in a week when the price per can is 70 cents, we set $PRICE1 = 70$. Using the linear regression model estimated in part (c), the point estimate for the expected sales is:

$$E(SAL1|\widehat{PRICE1} = 70) = b_1 + b_2(70)$$

Based on the regression output, the expected number of cans sold is approximately 10302.9 units.

Using statistical software to calculate the standard error of the expected value, the 90% interval estimate is:

$$[8624.934, 11980.87]$$

Conclusion:

We are 90% confident that when the price of brand no. 1 tuna is set to 70 cents, the average (expected) weekly sales volume will be between 8624.934 and 11980.87 cans.

- (e) Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 “at the means.” Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?

The price elasticity of demand for a linear model evaluated “at the means” is given by:

$$\epsilon = \beta_2 \left(\frac{\overline{PRICE1}}{\overline{SAL1}} \right)$$

Treating the sample means as constants, we first calculate the multiplier:

$$\frac{\overline{PRICE1}}{\overline{SAL1}} = 0.01164658$$

To construct the 95% interval estimate for the elasticity, we multiply the 95% interval bounds for β_2 (calculated in part c) by this constant multiplier:

$$\left[b_{2,\text{lower}} \left(\frac{\overline{PRICE1}}{\overline{SAL1}} \right), b_{2,\text{upper}} \left(\frac{\overline{PRICE1}}{\overline{SAL1}} \right) \right]$$

Using the regression output, the 95% interval estimate for the elasticity of sales is:

$$[-6.898149, -3.221501]$$

Economic Interpretation:

Because the entire interval falls below -1 (meaning the absolute value of the elasticity is strictly greater than 1), we find that sales of brand no. 1 tuna are **fairly elastic** with respect to price.

This makes perfect economic sense. Canned tuna is a highly substitutable good. If a supermarket raises the price of brand no. 1 tuna, consumers can easily switch to brand no. 2, brand no. 3, or even a different protein entirely. Because consumers have so many easy alternatives, a small percentage increase in price results in a much larger percentage drop in sales.

- (f) Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p-value for the test. What is your conclusion?

We are testing the hypothesis that the price elasticity of demand evaluated at the sample means (ϵ) is exactly -3 against the two-sided alternative. The elasticity is defined as $\epsilon = \beta_2 \left(\frac{\overline{PRICE1}}{\overline{SAL1}} \right)$.

To state the hypotheses strictly in terms of the model parameter β_2 , we isolate it:

$$H_0 : \beta_2 = -3 \left(\frac{\overline{SAL1}}{\overline{PRICE1}} \right)$$

$$H_1 : \beta_2 \neq -3 \left(\frac{\overline{SAL1}}{\overline{PRICE1}} \right)$$

Treating the sample means as constants, the standard error of our elasticity estimate is $se(\hat{\epsilon}) = \left(\frac{\overline{PRICE1}}{\overline{SAL1}} \right) se(b_2)$.

The test statistic is calculated as:

$$t = \frac{\hat{\epsilon} - \epsilon_0}{se(\hat{\epsilon})} = \frac{-5.06 - (-3)}{se(\hat{\epsilon})}$$

Using our regression output, the calculated test statistic is:

$$t = -2.250572$$

For a two-tailed test at the 10% level of significance ($\alpha = 0.10$) with $df = 52 - 2 = 50$, the critical t -value is $t_{0.05,50} = 1.676$. The rejection region is defined as $|t| \geq 1.676$, meaning we reject H_0 if $t \leq -1.676$ or $t \geq 1.676$.

The calculated p -value for this test is 0.02884204.

Conclusion:

Because our test statistic of -2.250572 falls into the rejection region (and the p -value is less than 0.10), we reject the null hypothesis. We conclude that at the 10% significance level, there is sufficient statistical evidence to claim that the price elasticity of demand for brand no. 1 tuna is different from -3 .